WIRESHARK SOC LAB

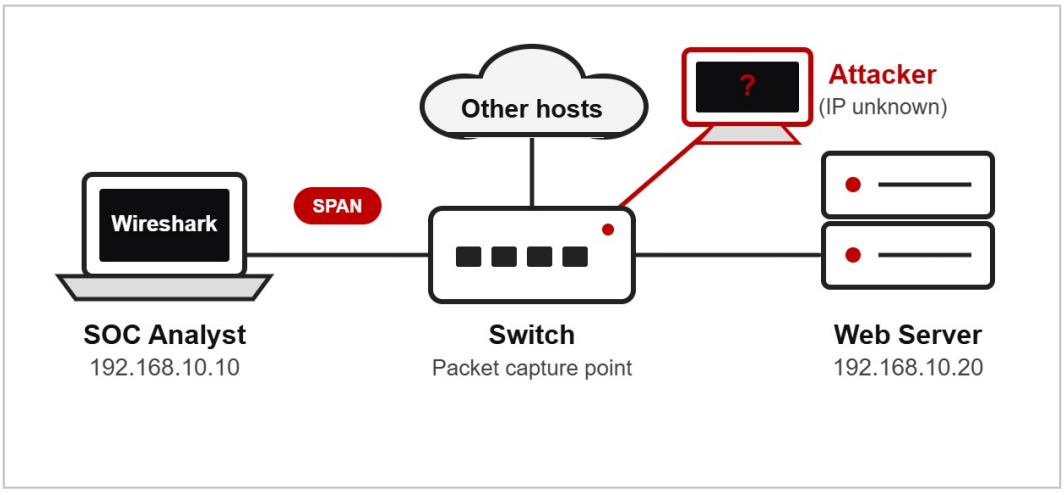
Identify the Attacker's IP

Online lab: https://networkwalks.com/wireshark-lab3/

SCENARIO

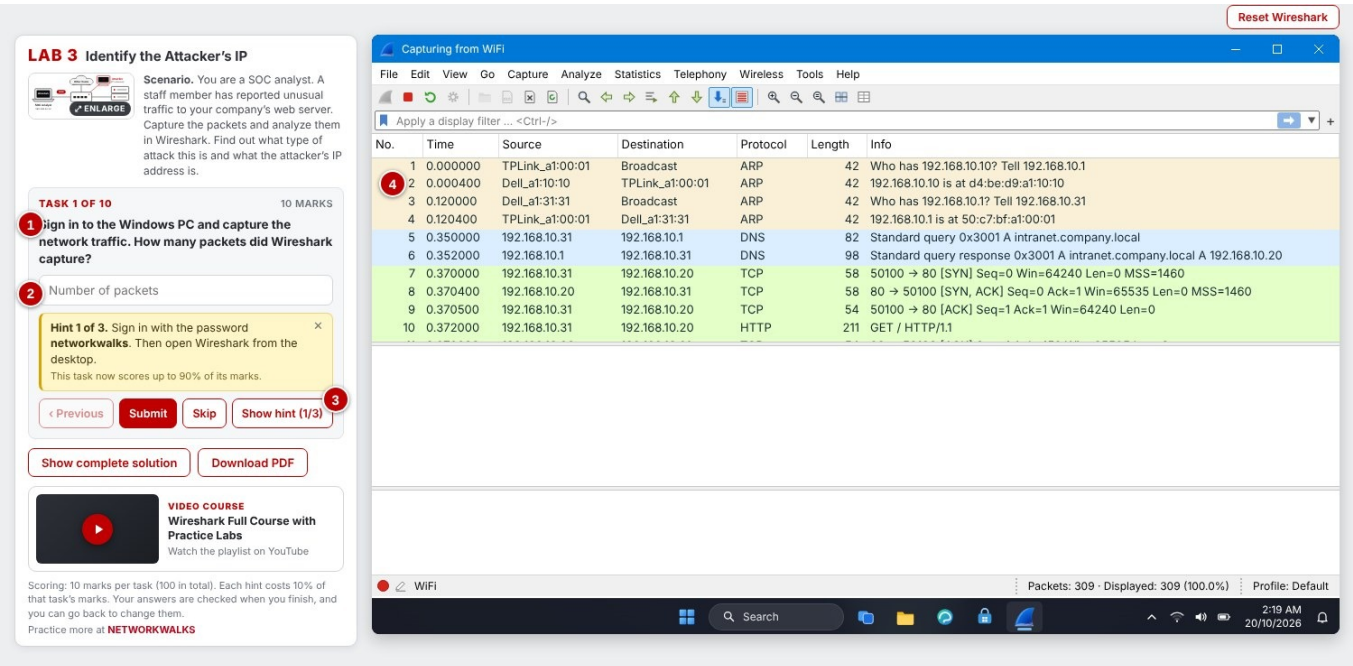
You are a SOC analyst. A staff member has reported unusual traffic to your company's web server. Capture the packets and analyze them in Wireshark. Find out what type of attack this is and what the attacker's IP address is.

TOPOLOGY DIAGRAM



LAB INSTRUCTIONS

How to complete this lab



The lab screen: the task box on the left, the Windows 11 PC with Wireshark on the right.

1. Read the scenario and the task in the left box (1). Do one task at a time.
2. On the right PC, sign in with the password hint. Double-click Wireshark, then double-click the WiFi interface to start the capture (4).
3. Use the filter bar, the packet list and the Statistics menu to find the answer for the task.
4. Type your answer in the box (2), then press Submit. Use Skip to leave a task for later and Previous to go back. Answers are checked at the end.
5. Stuck? Press Show hint (3). Each hint costs 10% of the marks of that task, only the first time you open it. A task with no answer scores 0.
6. On task 10 press Submit, add a comment if you like, then press Finish. You get a result card that you can download as a picture and share. Show complete solution is always available.

TASKS

Sign in to the Windows PC (user Student, the password hint is on the sign-in screen), start a capture in Wireshark and complete the 10 tasks below. Each task is worth 10 marks (100 in total). Each hint costs 10% of that task's marks.

Task 1. Capture the traffic

10 marks

Sign in to the Windows PC and capture the network traffic. How many packets did Wireshark capture?

Hint 1: Sign in with the password networkwalks. Then open Wireshark from the desktop.
Hint 2: Double-click the interface that shows an activity graph.
Hint 3: Read the number after Packets: in the status bar at the bottom of the Wireshark window.

Answer: 309

Solution: Sign in with the password networkwalks, double-click Wireshark on the desktop, then double-click the WiFi interface. The status bar shows Packets: 309.

Task 2. Show only SYN packets

10 marks

Most of this traffic is normal. Show only the packets that start a TCP connection (the SYN flag is set). How many packets are displayed?

Hint 1: Use the display filter bar. It is the green box below the toolbar.
Hint 2: Type tcp.flags.syn == 1 and press Enter.
Hint 3: Read the number after Displayed: in the status bar.

Answer: 73

Solution: Type

```
tcp.flags.syn == 1
```

in the filter bar and press Enter. The status bar shows Displayed: 73.

Task 3. Find the attacker

10 marks

Most PCs send only one or two of these packets. One IP address sends a long block of them, one after another. What is the attacker's IP address?

Hint 1: Type tcp.flags.syn == 1 in the filter bar and press Enter. Then scroll through the Source column.
Hint 2: Look for one IP that sends a long block of packets in a fraction of a second.
Hint 3: The block starts at about 3.4 seconds in the Time column.

Answer: 192.168.10.66

Solution: Type

```
tcp.flags.syn == 1
```

in the filter bar and scroll through the Source column. One source, 192.168.10.66, sends a block of about 40 packets from 3.4 seconds onward. That is the attacker.

Task 4. Find the target

10 marks

Which IP address is the target of the attacker's packets?

Hint 1: Type tcp.flags.syn == 1 in the filter bar and find the long block of packets sent by one IP address.
Hint 2: Look at the Destination column of the packets in that block.

Answer: 192.168.10.20

Solution: Type

```
tcp.flags.syn == 1
```

in the filter bar. The long block of packets comes from the attacker. The Destination column of those packets shows 192.168.10.20, the web server.

Task 5. Source port

10 marks

Look at the attacker's packets in the Info column. One port number is the same in every packet. What is it?

Hint 1: Type tcp.flags.syn == 1 in the filter bar and find the long block of packets sent by one IP address. Each line in the Info column starts with two port numbers with an arrow between them.
Hint 2: The number before the arrow is the source port. It is the same in every packet of that block.

Answer: 52311

Solution: Type

```
tcp.flags.syn == 1
```

in the filter bar and look at the attacker's packets. In the Info column, the number before the arrow is the source port. It is 52311 in every packet, while the destination port changes each time.

Task 6. Ports probed

10 marks

How many different ports did the attacker try on the target?

Hint 1: Open the Statistics menu and click Conversations.
Hint 2: Click the TCP tab. Find the attacker, 192.168.10.66, and its rows with 192.168.10.20.
Hint 3: Count those rows. Each row is one port.

Answer: 40

Solution: Open Statistics › Conversations and click the TCP tab. The attacker has 40 rows with 192.168.10.20, one for each port.

Task 7. Open ports

10 marks

Some of these ports are open on the server. An open port answers with a SYN, ACK packet. Which ports are open? Separate them with commas.

Hint 1: The attacker is 192.168.10.66. Type ip.addr == 192.168.10.66 in the filter bar and press Enter.
Hint 2: Look in the Info column for packets with [SYN, ACK]. They come from the server.
Hint 3: The number before the arrow in those lines is the open port.

Answer: 22, 80, 443

Solution: Type

```
ip.addr == 192.168.10.66
```

in the filter bar. The server answers [SYN, ACK] only from ports 22, 80 and 443. These ports are open.

Task 8. Reply for closed ports

10 marks

For all the other ports the server sends a different reply. Which flags does it send back for a closed port?

Hint 1: The attacker is 192.168.10.66. Type ip.addr == 192.168.10.66 in the filter bar and look at the replies that come from the web server, 192.168.10.20.
Hint 2: Most replies are not [SYN, ACK]. Check the flags shown in the Info column.

Answer: RST, ACK

Solution: Type

```
ip.addr == 192.168.10.66
```

in the filter bar. The other replies from the server show [RST, ACK]. The server uses a reset to say the port is closed.

Task 9. Closed ports

10 marks

How many ports are closed?

Hint 1: The attacker, 192.168.10.66, tried 40 ports. Three of them (22, 80 and 443) are open.
Hint 2: Closed ports = ports tried minus open ports.

Answer: 37

Solution: 40 ports were tried and 3 are open, so 37 ports are closed.

Task 10. Attack type

10 marks

What type of attack is this?

Hint 1: One IP tried many different ports in a very short time and sent no data.
Hint 2: A SYN flood sends thousands of packets to one port. This attacker tried many different ports to find which ones are open.

Answer: Port scan

Solution: One IP tried many ports in a very short time to find the open ones. This is a port scan.

SUMMARY

Attacker:	192.168.10.66
Target:	Web server 192.168.10.20
Ports tried:	40 (open: 22, 80, 443; closed: 37, reply RST, ACK)
Attack type:	Port scan

Watch the Wireshark video playlist: https://www.youtube.com/playlist?list=PLG-wUGWU39ro